

**ENGINEERING CHANGE NOTE (ECN)**

PROJECT: ProjectAsynTasking

ORIGINATION: Kid Kidd

PURPOSE OF CHANGE: Round-robin cyclic executive task manager results in a more reliable design. New design will reduce labor and design costs for implementing future extensions.

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**TEAM LEAD:** William Wang

**EXECUTIVE SUMMARY:**

The PROJECT-ASYNC design was modified to use a round-robin cyclic executive task manager. The LED and serial operations were separated into individual tasks. A function pointer array was added to dispatch these tasks in round-robin order. The customer and high-level technical requirements remain unchanged. Testing was extended to verify the operation of the cyclic executive.

**HIGH-LEVEL DESCRIPTION:**

Modify PROJECT-ASYNC to use a function-pointer-based round-robin cyclic executive task manager.

**REASON FOR CHANGE:**

Round robin cyclic executive task manager results in a more reliable design. New design will reduce labor and design costs for implementing future extensions.

**Changed Documents:**

[1] kiddKid.cmpe311.fall26.project#1 - 9/26/2026

The program design, logical flow chart, source code, and testing/validation requirements were modified to implement and verify the cyclic executive task manager.

**Added Documents:**

None

**Removed Documents:**

None

**Replaced Documents:**

None

**Any Additional Design Considerations:**

The cyclic executive dispatches the LED 1, LED 2, and serial input tasks through a function pointer array. Each task executes briefly and returns control so the next task can execute.